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Quantum Computing in Transportation Engineering: A Survey

Shriyank Somvanshi, Subasish Das, Md Monzurul Islam, Sazzad Bin Bashar Polock, Gaurab Chhetri, and Darrell Anderson

***Abstract*—Quantum computing introduces powerful computational paradigms capable of addressing optimization and machine learning challenges that are increasingly critical in transportation engineering. This paper presents a comprehensive survey of how quantum algorithms and models are being applied across transportation domains, including intelligent transportation systems, traffic management, vehicle routing, logistics, and autonomous systems. Unlike prior reviews that primarily introduce quantum theory, we systematically connect fundamental quantum principles-qubits, superposition, entanglement, and quantum gates-with specific transportation use cases. We synthesize recent findings showing that the Quantum Approximate Optimization Algorithm, Grover's search, and hybrid quantum–classical solvers can improve performance in combinatorial problems such as vehicle routing and dynamic traffic signal control, while quantum machine learning models (e.g., Quantum Convolutional Neural Networks, Quantum Generative Adversarial Networks, and clustering algorithms) demonstrate promise in traffic prediction, anomaly detection, and autonomous vehicle perception. Our review highlights that although quantum hardware limitations (noise, scalability, connectivity) remain significant, empirical studies already suggest measurable improvements in solution quality and computational efficiency when compared to classical heuristics. The key contribution of this survey is to map the landscape of quantum computing in transportation engineering, identify application-specific advantages and current bottlenecks, and outline future research directions toward scalable, domain-specific quantum algorithms. By doing so, we provide transportation researchers and practitioners with a structured foundation to integrate quantum approaches into emerging mobility systems, with implications for efficiency, safety, sustainability, and equity.**

***Index Terms*—Quantum computing, algorithms, transportation applications, Intelligent Transportation Systems (ITS), routing/logistics, optimization, machine learning.**

I. Introduction

TRANSPORTATION engineering encompasses several key domains, including traffic management, logistics, and infrastructure planning, which face significant computational challenges such as data scalability, real-time optimization, and system complexity [1]. The massive volume of data from moving objects, sensors, and queries in transportation systems requires advanced data management techniques [2]. Prior literature has addressed these challenges within specific contexts. For example, Arem [3] emphasizes equilibrium models and intelligent vehicle systems, identifying system complexity as a barrier to high-tech traffic systems. Del Ser et al. [4] discuss bioinspired computational intelligence for optimization in Intelligent Transportation Systems (ITS). Outay et al. [5] examine UAVs in road safety and infrastructure management. Ji et al. [6] review computational methodologies for infrastructure resilience, and Lana et al. [7] highlight data scalability and integration challenges in traffic forecasting. Across these domains, data-intensive applications strain conventional computational methods, underscoring the need for novel paradigms that can handle complexity, heterogeneity, and real-time constraints.

Quantum computing, leveraging principles like superposition and entanglement, has emerged as a transformative paradigm for optimization and machine learning (ML). These principles enable quantum systems to perform parallel computations and establish strong correlations among qubits, offering exponential speedups in specific scenarios. In transportation, this potential has drawn significant interest for computationally intensive tasks such as network analysis, traffic optimization, and route planning. For example, Cooper [8] explored the theoretical possibilities of quantum computing in transportation modeling. Wang et al. [9] demonstrated how quantum principles accelerate optimization and secure data transmission in ITS. Phillipson [10] highlighted their role in logistics optimization. Divya and Peter [11] showed the utility of quantum-enhanced machine learning for traffic flow analysis. Ajagekar and You [12] illustrated applications in energy systems optimization with implications for transportation. Wang et al. [13] examined Grover's search and Quantum Approximate Optimization Algorithm (QAOA) in engineering optimization, while Zhuang et al. [14] emphasized ITS applications such as traffic prediction and routing.

The rise of quantum computing presents a novel opportunity to transform transportation engineering by tackling complex computational challenges that conventional methods struggle to solve. Recent studies highlight promising areas such as shortest path computation, multi-objective routing, and net-

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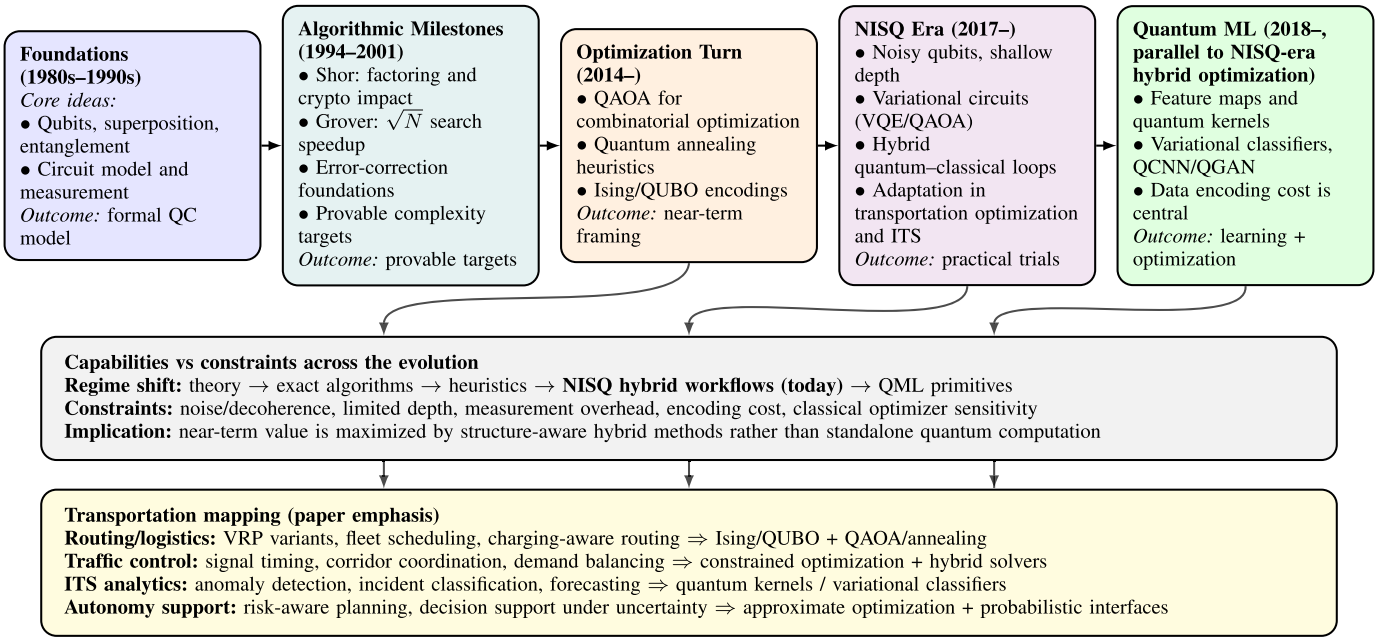


Fig. 1. Evolution of quantum computing from foundational theory to the NISQ era and quantum machine learning, showing why hybrid quantum–classical workflows dominate near-term practice. The bottom mapping summarizes how optimization- and learning-oriented QC methods align with transportation engineering problem classes discussed in this paper.

work optimization [8]. In ITS, quantum algorithms have shown potential in path planning and operations management by leveraging entanglement and superposition for large-scale data handling [9].

Fig. 1 illustrates the methodological evolution of quantum computing from foundational theory to NISQ-era hybrid optimization and quantum machine learning, highlighting how hardware constraints have shifted practical emphasis toward structure-aware hybrid workflows. The transportation mapping demonstrates how optimization- and learning-oriented quantum primitives align with routing, traffic control, and ITS analytics, motivating the near-term focus of this survey.

The main contributions of this paper are as follows:

- We provide a structured review of fundamental quantum computing concepts and algorithms relevant to transportation.
- We categorize and synthesize applications of quantum methods across traffic management, routing and logistics, security, and autonomous systems.
- We critically evaluate limitations, highlighting challenges in scalability, robustness, and domain integration.
- We propose future research directions, including benchmarks, hybrid models, digital twins, and embedding equity and sustainability in algorithm design.

To strengthen the literature review and explicitly highlight the research gap, we have added a comparative synthesis table (Table I). The table maps representative studies on quantum computing in transportation by summarizing their methods, application domains, contributions, and limitations. While prior works have addressed individual challenges such as routing, anomaly detection, or secure communication, they remain fragmented and domain-specific. Table I makes clear that no existing study provides an integrated, cross-domain assessment of quantum computing in transportation engineering. This gap underscores the novelty of our survey, which offers a structured synthesis that spans optimization, traffic management, anomaly detection, and secure communications. The next section builds upon this foundation by presenting detailed applications of quantum computing across transportation engineering.

Organization of the paper: To improve readability, this survey follows a dual framework. Section II introduces core principles of quantum computing. Section III consolidates algorithms and models (e.g., Grover's, Shor's, QAOA, quantum annealing, and QML). Section IV discusses applications in transportation domains including traffic management, routing/logistics, security, and autonomous systems. Section V provides a critical evaluation of research gaps, and Section VI concludes with future directions.

II. Quantum Computing Primer

This section is organized as: (1) Basic concepts (II-A); (2) Computational paradigms-gate model vs. annealing (II-B); (3) Core algorithms relevant to transportation (III); and (4) Quantum machine learning (IV). This explicit structure avoids overlap with the Applications and Challenges sections while keeping the technical background self-contained.

A. Basic Concepts

Quantum computing, utilizing the principles of quantum mechanics, solves problems that classical systems cannot. The foundational elements of this emerging technology comprise qubits, entanglement, superposition, and quantum gates. Collectively, these elements constitute the fundamental building blocks, offering exponential acceleration for specific problems such as optimization, cryptography, and simulation [14], [15].

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TABLE I
COMPARISON OF KEY REVIEWED PAPERS AND IDENTIFIED RESEARCH GAPS

Reference	Quantum Method / Model	Application Area	Key Contribution	Limitation / Gap
Cooper [8]	Fundamental QC principles, superposition/entanglement	Transport modeling	Theoretical exploration of QC potential in transportation	No empirical validation; does not extend to logistics or ITS deployment
Zhuang et al. [14]	Survey of QC in ITS	Traffic prediction, routing	Comprehensive survey of quantum methods for ITS	Focused narrowly on ITS; limited coverage of logistics, sustainability, or AVs
Osaba et al. [15]	QC for routing, optimization	Vehicle routing problems (VRP)	Systematic review of QC for routing optimization	Does not address broader transportation engineering domains
Harikrishnakumar et al. [16]	QAOA, quantum annealing	Bike-sharing, VRP	Compared solvers for transport-related optimization	Hardware scalability challenges; limited real-world validation
Jagirdar [17]	Quantum annealing	Traffic flow, logistics	Highlighted QC for mobility and logistics	High-level discussion; lacks detailed algorithmic mapping
Gautam & Ahn [18]	Quantum Path Integral Approach (QPIA)	Vehicle routing	Novel QC-inspired heuristic for constrained VRP	Only proof-of-concept; limited scalability analysis
Rattan et al. [19]	QML (QSVM, QNNs)	AVs, ADAS	Reviewed QML models for autonomous systems	Emerging area; lacks empirical large-scale transport datasets
Zhang et al. [20]	Hybrid quantum genetic algorithms	Traffic flow forecasting	Demonstrated QGA for congestion reduction	Limited robustness under adversarial/real-time conditions
Saxena & Nigam [21]	Hybrid QC-classical anomaly detection	ITS anomaly detection	Applied quantum kernels for small datasets	Limited to controlled experiments; scalability and generalization unclear
Wang et al. [9]	Quantum gates, circuits	Cryptography, secure V2V/V2I	Showed QC impact on security and optimization in transport	Hardware noise and decoherence challenges; security solutions remain conceptual
Azad et al. [22]	Grover's algorithm	Route planning, search	Applied Grover's search for faster route optimization	Focus only on small datasets; lacks hybrid scalability analysis
Alsaiyari et al. [23]	Variational QC algorithms	Secure ITS communications	Applied variational solvers to V2I encryption	Early-stage application; limited empirical results
Childs et al. [24]	Quantum random walks	Graph traversal, routing	Demonstrated exponential speedup for traversal problems	Theoretical; transport applications remain speculative
Venegas-Andraca [25]	Quantum random walk cryptography	Network security	Linked quantum walks with cryptography frameworks	Focused on theory; not applied in transport-specific settings

1) Qubits: A qubit, or quantum bit, serves as the quantum counterpart to a classical bit, the basic unit of information in traditional computing. Unlike classical bits, which can only take on discrete values of 0 or 1, qubits leverage quantum superposition to exist in multiple states at once. This unique characteristic allows quantum computers to execute numerous computations simultaneously, significantly increasing their computational capabilities [8]. The state of a qubit can be described mathematically as a linear combination (superposition) of two basis states, $|0\rangle$ and $|1\rangle$, as follows:

$$|\psi\rangle = \alpha|0\rangle + \beta|1\rangle$$

In this context, the complex probability amplitudes, denoted by α and β, are expressed as the square of their respective values, which quantifies the probability of measuring the qubit in either of the two possible states:

$$|\alpha|^2 + |\beta|^2 = 1$$

Ensuring that the total probability of all possible outcomes equals 1 is the purpose of this normalization condition [16].

2) Superposition: In quantum mechanics, superposition enables a single qubit to simultaneously exist in a combination of both the 0 and 1 states. To illustrate, consider a qubit in a state of:

$$|\psi\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

which has equal probabilities (50%) of collapsing to either state upon measurement. This property allows quantum algorithms to explore multiple candidate solutions in parallel, enabling speedups for search and optimization tasks [15].

3) Quantum State Visualization: To intuitively understand a qubit's behavior, it is often visualized using the Bloch sphere as illustrated in Fig. 2. The Bloch sphere is a geometric representation where:

- The state $|0\rangle$ associated with the north pole.
- The state $|1\rangle$ associated with the south pole.
- A valid qubit state represents any point on the surface of the sphere.

The coordinates of a point on the Bloch sphere are determined by the angles θ and ϕ, where the state can be represented by:

$$|\psi\rangle = \cos\left(\frac{\theta}{2}\right)|0\rangle + e^{i\phi}\sin\left(\frac{\theta}{2}\right)|1\rangle$$

This visualization aids in understanding how qubits can be manipulated through quantum gates, which rotate the qubit's state on the Bloch sphere [9], [14]. Beyond serving as a geometric illustration, Fig. 2 highlights how qubits encode superposition and phase information in a compact form. This is particularly important because operations such as the Hadamard or Pauli-X gate correspond to rotations on the sphere, making the figure a practical tool for reasoning about quantum algorithms [26]. In the context of transportation, the

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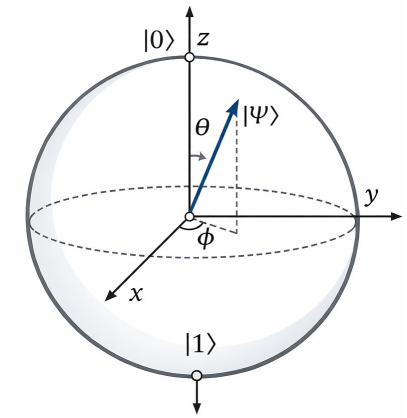


Fig. 2. Bloch Sphere.

Bloch sphere helps conceptualize how quantum algorithms like QAOA explore multiple routing configurations simultaneously through state rotations, which would be difficult to capture with equations alone. Thus, the figure not only represents the mathematics of qubits but also demonstrates the mechanism by which quantum gates manipulate states for real-world applications.

4) Entanglement: Two or more particles in quantum mechanics can become entangled, causing any change in one to immediately impact the others, no matter how far apart they are. This is known as quantum entanglement, and it's a concept that challenges our understanding of physics [14]. Entanglement is essential for quantum technologies, driving advancements in cryptography, quantum computing, and communication [9].

a) Mathematical representation of entangled states: Entangled states are described by a single wavefunction representing the entire system rather than the independent states of individual particles. For two qubits, an entangled state, such as a Bell state, is mathematically expressed as:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|00\rangle + |11\rangle)$$

Here, $|00\rangle$ and $|11\rangle$ are the basis states of the system. In this state, if the first qubit is measured to be $|0\rangle$, the second qubit will instantly collapse to $|0\rangle$, and the same applies to $|1\rangle$. This immediate correlation persists regardless of the distance between the two qubits, showcasing the non-local nature of entanglement [8], [17].

b) Non-locality and correlations: Entanglement demonstrates the phenomenon of non-locality, meaning that the measurement outcomes of entangled particles exhibit strong correlations, violating classical concepts of locality. These correlations have been verified by tests of Bell's theorem, proving that a local hidden-variable theory cannot fully explain the observed results [14]. For instance, consider two qubits in the entangled state:

$$|\psi\rangle = \frac{1}{\sqrt{2}}(|01\rangle + |10\rangle)$$

Measuring one qubit as $|0\rangle$ instantly puts the other in the $|1\rangle$ state, and vice versa. These correlations are key to quantum protocols such as quantum cryptography and quantum teleportation [16].

c) Physical implementation of entanglement: Various physical systems can be used to generate and maintain entanglement, including photons, trapped ions, and superconducting qubits. Spontaneous parametric down-conversion creates entanglement in photonic systems: a high-energy photon splitting into two lower-energy entangled photons. In ion traps, quantum gates like the Controlled-NOT (CNOT) gate are employed to entangle qubits by manipulating their quantum states. Superconducting qubits in IBM and Google quantum computers entangle electrons through precise control of electric current and magnetic fields in quantum circuits [9], [15].

Qubits, superposition, and quantum gates form the backbone of quantum computing. By enabling parallel computations through superposition and exploiting quantum mechanical phenomena like entanglement and interference, quantum computing offers transformative potential across industries, including transportation, logistics, and cryptography. The visualization of qubits using the Bloch sphere enhances comprehension of quantum state manipulation, making it an essential tool in quantum algorithm design [17].

B. Quantum Computing Paradigms

1) Gate-Based Quantum Computing: Gate-based quantum computing relies on quantum circuits built from quantum gates, which function similarly to classical logic gates but manipulate qubits. These models leverage entanglement, superposition, and interference to execute computations beyond the capabilities of classical computers. Each quantum gate represents a reversible operation, mathematically expressed as a unitary matrix, which transforms the quantum state of one or more qubits [14]. Gate-based models are key to quantum algorithms like Grover's and Shor's, which speed up computing significantly compared to classical methods [8].

a) Quantum gates and operations: Quantum gates perform operations on qubits, which can be represented by vectors in a Hilbert space. Some common gates include:

- **Pauli-X gate (X):** Flips the state of a qubit, analogous to a classical NOT gate:

$$X = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

- **Hadamard gate (H):** Places a qubit into an equal superposition of basis states:

$$H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$$

- **Controlled-NOT (CNOT):** A two-qubit gate that flips the second qubit if the first qubit is in state $|1\rangle$ [14].

The quantum circuit model is built by applying a sequence of these gates, transforming the input qubits through unitary operations. The final state is then measured, collapsing the quantum system into one of the basis states with a probability defined by the Born rule.

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b) Mathematical foundation: The state of a quantum system with n qubits is represented by a 2^n-dimensional vector. Quantum gates act on these states according to the Schrödinger equation:

$$i\hbar \frac{d}{dt}|\psi(t)\rangle = H|\psi(t)\rangle$$

Here, H is the Hamiltonian representing the total energy of the system. In gate-based quantum computing, the system evolves through discrete, unitary transformations [16].

c) Applications and significance: Gate-based quantum computing models have demonstrated their potential to solve various optimization, search, and cryptographic problems. At a high level, these capabilities arise from coherent interference and entanglement that shape probability amplitudes toward desirable solutions while preserving reversibility and unitary evolution [8]. A practical limitation is sensitivity to noise and decoherence in Noisy intermediate-scale quantum (NISQ) computing devices, which constrains circuit depth and motivates error-mitigation and compilation strategies tailored to hardware topology [17].

2) Quantum Annealing: Quantum annealing is an optimization method that harnesses quantum mechanics to tackle complex problems by identifying the system's lowest energy state. This process is based on adiabatic quantum computation principles, where the system starts in a known ground state and gradually evolves to an optimal solution [14]. Unlike gate-based quantum computing, which manipulates qubits through sequential operations, quantum annealing continuously transforms the energy landscape of qubits, making it particularly effective for combinatorial problems like the Vehicle Routing Problem (VRP) and the Traveling Salesman Problem [15].

Mathematically, quantum annealing can be expressed through the time-dependent Hamiltonian:

$$H(t) = A(t)H_{\text{initial}} + B(t)H_{\text{final}}$$

where H_{initial} represents the initial Hamiltonian with a known ground state, H_{final} encodes the problem's objective function, and $A(t)$ and $B(t)$ control the interpolation between these Hamiltonians over time. The system evolves under the Schrödinger equation, which governs the probability amplitude distributions across possible solutions [18].

One of the most common formulations in quantum annealing is the *Quadratic Unconstrained Binary Optimization (QUBO)* model. This model defines an optimization problem as:

$$\text{Minimize:} \quad x^T M x + c$$

subject to $x \in \{0, 1\}^n$, where M is an $n \times n$ matrix and c is a scalar offset. The Hamiltonian corresponding to this problem can be converted to an Ising model to facilitate solution finding on quantum annealers [16]. In practice, embedding large QUBOs onto sparse hardware graphs introduces chain-length and precision trade-offs that can affect solution quality and runtime, motivating hybrid pre-/post-processing to manage scale and noise [8], [9].

3) Quantum Machine Learning: Quantum Machine Learning (QML) is the combination of quantum computing and machine learning techniques to improve data processing, pattern recognition, and optimization capabilities. Using quantum principles such as superposition and entanglement, QML algorithms can process multiple states simultaneously. This enables faster and more efficient solutions to complex transportation problems. Quantum versions of machine learning models-such as quantum convolutional neural networks (QCNNs), quantum generative adversarial networks (QGANs), and quantum clustering algorithms-have emerged as key advances in this field.

4) Comparative Analysis With Classical Approaches: Classical computational methods have played a pivotal role in addressing transportation optimization problems, leveraging heuristic and metaheuristic algorithms (e.g., Genetic Algorithms, Ant Colony Optimization), machine learning models (e.g., RNNs, ARIMA), and mathematical programming techniques (e.g., Mixed-Integer Linear Programming (MILP)). These methods are particularly effective for medium-scale problems but often face scalability challenges when applied to large, real-time datasets due to high computational complexity [27].

In recent years, several advanced nature-inspired metaheuristics have emerged to improve solution quality and convergence speed for complex engineering optimization tasks. The Manta Ray Foraging Optimization (MRFO) algorithm [28] exemplifies this trend, using biologically inspired cyclone, chain, and somersault foraging strategies to achieve a strong balance between exploration and exploitation. Similarly, the Chaotic Quasi-oppositional Farmland Fertility Algorithm [29] integrates chaotic mapping and opposition-based learning to overcome local optima and accelerate convergence, outperforming classical methods on a suite of benchmark and real-world problems. Another recent development is the Beluga Whale Optimization algorithm [30], which models collective whale foraging behavior and has rapidly expanded through numerous improvement and hybridization studies, demonstrating high adaptability across diverse optimization domains.

Beyond algorithmic development, metaheuristics have been increasingly applied to optimize distributed computing environments that underpin modern ITS. In particular, bio-inspired algorithms have shown promise for dynamic task offloading across cloud, fog, and edge computing networks, improving resource utilization, reducing latency, and balancing energy consumption in complex IoT settings [31]. These approaches parallel the computational demands of connected and autonomous vehicle (AV) networks, where low-latency decision-making is critical.

Bridging classical and quantum paradigms, quantum-inspired metaheuristic algorithms have emerged as a promising intermediate category. These methods incorporate quantum principles such as superposition and parallelism to enhance classical metaheuristics, yielding faster convergence and improved search diversity for complex combinatorial problems [32]. Quantum inspiration has been successfully integrated into evolutionary and swarm-based frameworks, providing a stepping stone toward fully quantum optimization in transportation and logistics contexts.

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III. Quantum Computing in Transportation Engineering

1) Quantum Algorithms:

Quantum algorithms have become powerful tools for addressing computationally demanding problems in diverse fields, including transportation. Algorithms like Grover's and Shor's provide critical advantages by improving the speed of search operations and the efficiency of cryptographic security systems, respectively [9], [22]. These enhancements have applications in route planning, congestion management, and secure communication protocols in ITS [23]. Additionally, quantum optimization methods, such as the QAOA, and techniques based on Quantum Random Walks, demonstrate great potential in tackling NP-hard transportation challenges like the VRP [33]. By utilizing quantum principles of superposition and entanglement, these techniques enable scalable, real-time solutions that could redefine transportation infrastructure and logistics.

a) Grover's algorithm: Grover's algorithm is a quantum search algorithm that finds unstructured data sets faster than classical methods. While classical search algorithms require $O(N)$ operations to find a solution in a data set of size N, Grover's algorithm accomplishes the task in $O(\sqrt{N})$. The algorithm uses Grover iteration, with two main steps: the oracle O and the diffusion operator D. The oracle reverses the phase of the target state:

$$O|x\rangle = \begin{cases} -|x\rangle, & \text{if } x = x_{\text{target}} \\ |x\rangle, & \text{otherwise} \end{cases}$$

The diffusion operator amplifies the amplitude of the target state by reflecting around the average amplitude:

$$D = 2|\psi\rangle\langle\psi| - I$$

By applying these operations approximately $\frac{\pi}{4}\sqrt{N}$ times, Grover's algorithm maximizes the probability of measuring the correct state. In transportation, Grover's algorithm can be applied to optimize route searches, traffic flow analysis, and real-time data queries in ITS [9], [22].

This algorithm can enhance real-time data processing by speeding up unstructured search operations, enabling faster route planning and dynamic traffic management in large urban networks. Its quadratic speedup in search tasks allows ITS to quickly analyze large datasets for better traffic control and resource allocation

b) Shor's algorithm: Shor's algorithm efficiently factors large integers, breaking cryptographic systems based on RSA encryption; classical factoring algorithms have exponential time complexity while Shor's reduces this to polynomial time using quantum Fourier transforms. The algorithm operates in two phases:

1) **Quantum Period Finding**: Shor's algorithm finds the period r given a function $f(a) = x^a \mod N.$, where $f(a + r) = f(a)$.
2) **Classical Post-Processing**: Once the period r is known, factors of N can be determined using the relationship:

$$\gcd(x^{r/2} - 1, N) \quad \text{or} \quad \gcd(x^{r/2} + 1, N)$$

The quantum component involves preparing a superposition of states and applying the quantum Fourier transform:

$$QFT|a\rangle = \frac{1}{\sqrt{N}} \sum_{k=0}^{N-1} e^{2\pi i a k/N} |k\rangle$$

Shor's algorithm is relevant to transportation because it highlights the vulnerability of classical encryption methods used in vehicle-to-infrastructure (V2I) and vehicle-to-vehicle (V2V) communications. With the growing adoption of autonomous and connected vehicles, secure data transmission is essential to prevent cyberattacks and maintain operational integrity [23].

Shor's algorithm can accelerate the shift towards quantum-resistant cryptography by efficiently factoring large numbers, posing a threat to classical cryptographic systems. This has prompted the development of post-quantum security protocols for critical communication networks in transportation, such as vehicle-to-vehicle (V2V) and vehicle-to-infrastructure (V2I) communications [8].

These algorithms have also inspired hybrid quantum-classical approaches, such as the *QAOA*, which addresses complex routing and scheduling problems. For example, QAOA has demonstrated significant improvements in solving combinatorial optimization tasks like the VRP, a critical challenge in logistics and transportation planning [16].

2) Quantum Optimization Techniques:

Quantum optimization techniques, including quantum annealing, the QAOA, and the Quantum Path Integral Approach (QPIA), offer promising solutions to complex combinatorial optimization problems in transportation and logistics. Quantum annealing evolves a system towards its lowest-energy state to find optimal solutions efficiently, demonstrating applications in network optimization and vehicle routing [14], [15]. Similarly, QAOA employs hybrid quantum-classical iterations to refine solutions, making it suitable for tasks like rebalancing bike-sharing systems and dynamic resource allocation [16]. The QPIA method, inspired by Feynman's path integral formalism, evaluates all potential routes using quantum interference and has shown potential for optimizing constrained VRP [18]. These techniques enhance transportation modeling by improving route planning, traffic management, and infrastructure resource placement, such as electric vehicle charging stations. While current quantum hardware limitations, such as qubit noise, pose challenges, ongoing research into hybrid algorithms and error correction aims to make these optimization approaches increasingly viable [16]. Across these families, performance depends on problem encoding quality (cost/constraint maps), hardware-aware compilation or embedding, and robust classical post-processing-considerations that directly influence scalability and solution fidelity on present-day devices [9], [34].

3) Quantum Random Walks:

Quantum random walks extend the concept of classical random walks by utilizing quantum principles such as superposition and interference. In a classical random walk, a particle

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moves through nodes in a graph or along a line, randomly choosing its next position at each step. Over time, the particle's position distribution becomes unimodal, typically centered near the starting point. In contrast, quantum random walks enable the particle to exist simultaneously at multiple nodes, creating a probability distribution influenced by quantum interference, which often results in a bimodal pattern. This distribution has peaks near both the origin and a distant location, with lower probabilities at intermediate positions where destructive interference occurs [8]. Key performance metrics in random walks include the *mix time* and the *hit time*. The mix time refers to the duration required for the particle's position distribution to converge to a long-term average, while the hit time measures how long it takes to reach a specific node. Quantum random walks can achieve at most a quadratic speedup in mix times compared to classical walks, although their performance can vary depending on the graph's structure [35]. On certain complex graphs, such as hypercubes, quantum hit times can be exponentially faster than classical counterparts [24]. These speed improvements make quantum random walks particularly valuable in solving graph-based problems efficiently. For example, quantum walks can accelerate algorithms for graph traversal and state-space exploration. They have also demonstrated potential for enhancing optimization algorithms, where nodes on a graph represent possible states of a system. Problems requiring the identification of specific states analogous to hitting a target node can benefit from the reduced hit times offered by quantum random walks [8]. In practical applications, quantum random walks have implications for both quantum algorithm design and cryptography. They provide a foundation for certain quantum search algorithms and optimization techniques, enabling faster solutions to problems related to traffic flow, network optimization, and state transition analysis in transportation systems [8]. Moreover, Quantum walks challenge classical cryptography, which relies on the difficulty of graph problems. This has led to the development of post-quantum cryptography methods [25]. Overall, quantum random walks offer a significant theoretical advantage for many computational tasks. While their efficiency gains are context-dependent, ongoing research continues to explore ways to use these capabilities across various fields, including transportation modeling and intelligent systems [24].

IV. Quantum Computing Applications in Transportation Engineering

A. Intelligent Transportation Systems

Real-time traffic flow prediction, congestion forecasting, and adaptive signal control are challenged by high-dimensional, non-Euclidean network structure and temporal dependencies; Spatial-Temporal Graph Convolutional Networks (STGCNs) address this, and quantum-enhanced models further improve computational efficiency, accuracy, and adaptivity. Traditional traffic prediction methods often struggle with the high-dimensional, non-Euclidean structure of traffic networks and the inherent temporal dependencies in transportation data. STGCNs have emerged as a promising solution, leveraging graph-based deep learning to model complex traffic dynamics. However, quantum-enhanced STGCNs and quantum optimization techniques offer further improvements in computational efficiency, predictive accuracy, and real-time adaptability.

QML for perception and anomaly detection. Quantum versions of machine learning models (e.g., QCNNs, Quantum Support Vector Machines (QSVMs)) can accelerate perception tasks in ITS by exploiting superposition and entanglement for parallel feature processing, aiding real-time camera/sensor streams and congestion prediction [14], [19], [36]. Quantum spatial–temporal models and hybrid quantum–classical detectors have also shown promise for robust anomaly detection under adversarial conditions [9], [21].

Recent studies have explored different facets of this integration. Qu et al. [37] introduced a Temporal-Spatial Quantum Graph Convolutional Neural Network (TS-QGCNN), which applies Schrödinger-based quantum modeling to capture both temporal and spatial dependencies in traffic congestion forecasting. By leveraging quantum graph convolutional networks (QGCNNs), this approach enhances the ability to predict congestion patterns in complex, non-Euclidean traffic networks, addressing one of the core challenges in ITS. Meanwhile, Moskvin [38] highlighted the potential of Quantum Neural Networks (QNNs) in traffic prediction by utilizing quantum superposition, entanglement, and interference to efficiently process large-scale transportation datasets. Although QNNs improve traffic flow forecasting, their application in real-time adaptive control and integration with STGCNs remains an open research area.

Annealing-based deployments in ITS. Quantum annealing has been used for domain-specific ITS tasks such as real-time traffic-flow optimization and resource allocation; these deployments complement data-driven prediction by enabling rapid, combinatorial decision-making under congestion and demand variability [16], [17].

In addition to quantum-enhanced deep learning models, quantum optimization techniques have been explored to refine prediction and control mechanisms. Zhang et al. [20] applied Quantum Particle Swarm Optimization (QPSO) to optimize neural network hyperparameters, significantly reducing prediction errors in traffic flow forecasting. Beyond neural network optimization, quantum-based methods have shown promise in adaptive traffic signal control and congestion forecasting. Singh et al. [39] proposed a quantum annealing-based approach to traffic control using QUBO, which dynamically adjusts the timing of traffic signals based on real-time conditions and significantly reduces traffic congestion in high-traffic urban scenarios. Similarly, Marchesin et al. [40] introduced a versatile quantum annealing model for urban traffic mobility optimization, leveraging D-Wave's quantum machines to optimize real-time traffic light activation patterns and reduce congestion by considering traffic density at intersections. Furthermore, Acampora et al. [41] demonstrated the effectiveness of hybrid quantum-classical genetic algorithms for optimizing traffic signal timings in Network Signal Setting Design, reducing intersection delays through a Quantum Mating Operator that enhances evolutionary optimization efficiency.

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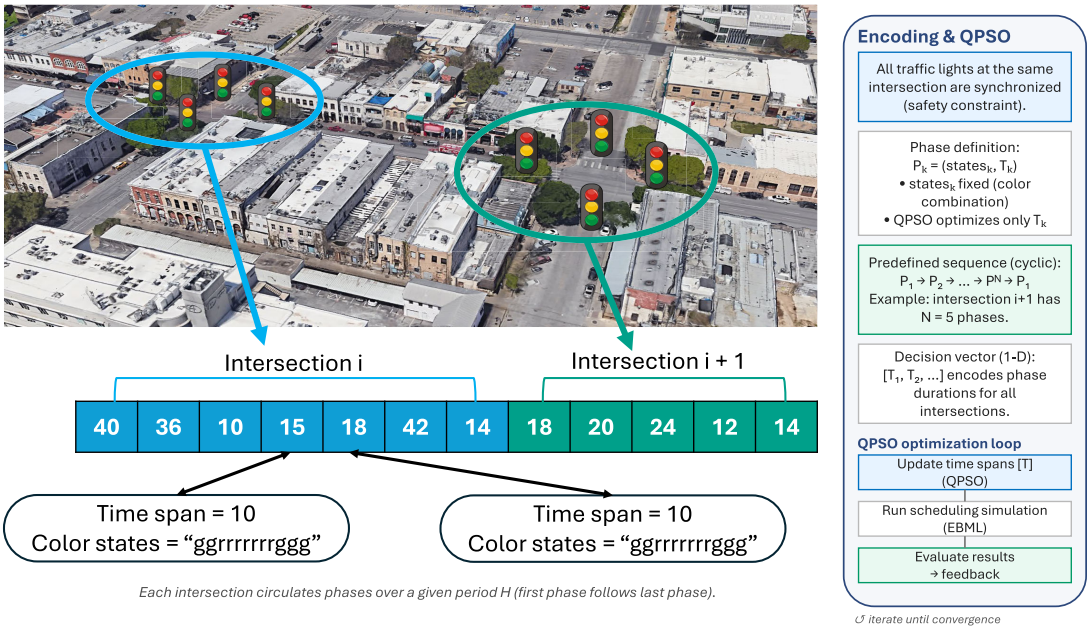


Fig. 3. Traffic Light Scheduling using QPSO, Adapted from [42]. Traffic signal phases follow a fixed cyclic sequence, while QPSO optimizes the corresponding phase durations across intersections under synchronization and safety constraints.

B. Traffic Management

The optimization of real-time traffic signal control is an essential challenge in urban transportation systems, requiring adaptive methodologies to address fluctuating traffic demands. Classical Particle Swarm Optimization (PSO) has been widely utilized for traffic signal optimization, leveraging its ability to explore large search spaces efficiently [43]. However, classical PSO faces limitations in complex and real-time traffic scenarios, often struggling with slow convergence and the risk of getting trapped in local optima, making it suboptimal for highly dynamic urban environments [42].

Recent advancements in QPSO have demonstrated superior optimization performance over classical PSO, particularly in scenarios requiring faster convergence and better global search capabilities. QPSO has been successfully applied to urban traffic light scheduling, showing improvements in adaptive signal control and congestion reduction [42]. Fig. 3, QPSO encodes the timing schedules of traffic signals across multiple intersections into a one-dimensional time-span vector, optimizing phase transitions for improved traffic flow. Each intersection follows a cyclic phase transition, where the first phase follows after the last, ensuring smooth coordination across intersections. By integrating quantum wave mechanics, QPSO mitigates the premature convergence issues of traditional PSO, leading to enhanced global search capability and better synchronization of traffic signals. In practice, this is particularly valuable for dense urban corridors such as central business districts, where traffic queues at one intersection can quickly spill over to the next. QPSO-based signal optimization enables corridor-level coordination, reducing stop-and-go traffic patterns and minimizing delays during peak commuting hours.

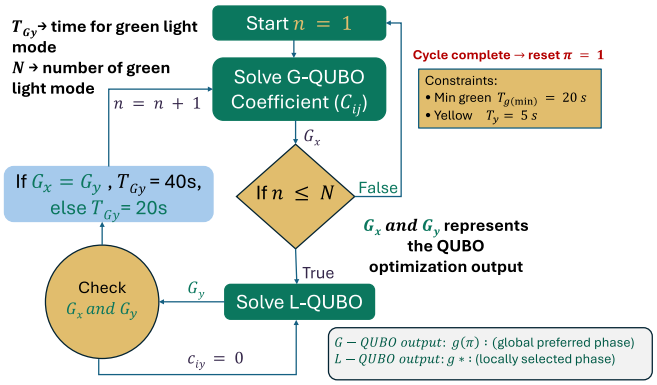


Fig. 4. Flowchart of the proposed two-stage QUBO traffic-signal control with fair sharing, adapted from [39]. G-QUBO selects the global preferred phase and L-QUBO refines the selection to prevent repetition under unbalanced demand. Green time is set to 40 s for matched selections and 20 s otherwise, with a fixed 5 s yellow, until the cycle completes and resets.

In Fig. 3, each block represents the allocated green or red light duration at an intersection, with color states (e.g., "ggrrrrrrggg") denoting the sequence of phases. The optimization process adjusts these spans simultaneously for multiple intersections (i and i+1 in the example), ensuring that phase transitions remain coordinated across the network. This allows traffic flow to be smoothed at a corridor level, reducing queue spillovers from one intersection to the next. Compared to classical PSO, QPSO explores a larger solution space due to quantum-inspired position updates, which helps avoid local optima and yields schedules that minimize overall delay and congestion. This makes the approach particularly effective for dense urban networks where inter-signal coordination is critical.

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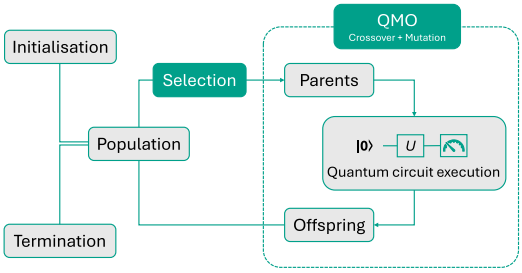


Fig. 5. Hybrid Quantum-Classical Genetic Algorithm Workflow with QMO Operations, Adapted from [41].

The current applications of QPSO have been largely concentrated on traffic flow forecasting, with limited exploration of its potential for direct real-time signal optimization. A study by Zhang et al. [20] has demonstrated QPSO's efficacy in improving traffic flow prediction by integrating it with radial basis function neural networks and genetic simulated annealing, leading to enhanced forecasting accuracy. In contrast, Quantum Annealing, particularly via QUBO, has shown significant advantages for real-time traffic signal control. Singh et al. [39] and Enan et al. [44] applied QUBO-based models to optimize signal timing dynamically, demonstrating superior performance compared to traditional fixed-cycle methods, especially under unbalanced traffic loads. Singh et al. [39] introduced a two-stage QUBO framework (Fig. 4) to ensure fair selection of green light phases within a cycle. This approach combines a Global QUBO (G-QUBO) to determine an initial green phase and a Local QUBO (L-QUBO) to refine phase transitions, preventing phase dominance in scenarios with highly unbalanced traffic.

As illustrated in Fig. 4, the two-stage framework first solves a global QUBO (G-QUBO) to select the globally preferred signal phase, followed by a local QUBO (L-QUBO) that refines the selection to prevent repeated phase activation under unbalanced demand. The control logic iterates across N phases within each cycle group, enforcing fairness through temporary coefficient suppression and resetting once the cycle completes. Green time is dynamically adjusted-extended to 40 s when global and local selections align, or set to a minimum of 20 s otherwise-while maintaining a fixed 5 s yellow phase. This flow highlights how fairness and operational constraints are directly embedded within the QUBO formulation to enable adaptive signal timing under varying traffic conditions. Related extensions, such as quantum annealing–supported virtual traffic lights [44], further demonstrate the potential of quantum-enhanced traffic control beyond classical rule-based strategies.

Beyond QPSO, other quantum-enhanced metaheuristics, such as Quantum Genetic Algorithms and Quantum-Inspired Firefly Algorithm (QIFA), have further advanced traffic signal optimization. Quantum Genetic Algorithms have been applied to network signal setting design, achieving better green signal timing optimization by leveraging quantum crossover and mutation operations executed on a quantum device [41]. As illustrated in Fig. 5, this hybrid quantum-classical genetic algorithm evolves binary chromosomes representing stage durations and junction offsets, with the Quantum Mating Operator performing crossover and mutation within a quantum circuit, while selection and other evolutionary processes remain classical. The workflow begins with initialization and population generation, followed by classical selection of parent chromosomes. These parents are then passed to the quantum circuit, where quantum gates implement crossover and mutation (QMO), producing offspring with higher diversity. The offspring rejoin the evolutionary loop for further selection and evolution, with termination reached once optimization criteria are satisfied. By embedding quantum operations into the genetic cycle, the algorithm reduces premature convergence and explores a wider solution space compared to purely classical GAs. This approach has been specifically used to minimize total deterministic delay across interacting junctions, directly impacting vehicle waiting times and improving traffic flow efficiency. Meanwhile, QIFA has enhanced adaptive signal scheduling by integrating quantum-inspired search strategies with firefly-based heuristics, optimizing phase transitions dynamically [45]. These quantum-enhanced methods highlight the growing potential of quantum computing in overcoming the computational constraints of classical optimization techniques for real-time traffic management.

QNNs and quantum graph models report gains in traffic prediction and anomaly detection by leveraging parallel state processing and interference [9], [20], [21], [46]. Hybrid quantum–classical designs are promising under NISQ noise, with empirical improvements on urban datasets; detailed deployments are developed in the Applications section. In practice, anomaly detection is essential for ensuring resilience in ITS. For example, sudden multi-vehicle accidents on freeways, unplanned lane closures during road work, or faulty loop detectors at intersections can create abnormal traffic patterns that classical models may fail to flag quickly. Quantum-enhanced models such as QSVMs and QCNNs accelerate feature extraction from high-dimensional sensor and camera data streams, enabling earlier identification of such anomalies. Similarly, in connected vehicle networks, quantum kernels can detect irregular V2V or V2I communication patterns that may indicate cyberattacks or malfunctioning units. These realistic use cases demonstrate how quantum-assisted anomaly detection can support faster incident response, maintain traffic stability, and improve overall system safety.

C. Vehicle Routing, Logistics, and Autonomous Systems

Quantum computing has emerged as a transformative technology with the potential to revolutionize complex optimization problems in transportation. One of its most promising applications lies in vehicle routing, where classical algorithms often struggle to solve large-scale combinatorial optimization problems efficiently. Studies have demonstrated that quantum computing, particularly through quantum variational algorithms and QUBO formulations, can significantly enhance VRP solutions by reducing computational complexity and execution time [27]. For instance, in large-scale logistics networks where hundreds of delivery trucks must be assigned to time-constrained routes, classical solvers face difficulties

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in recomputing assignments when disruptions such as traffic congestion or road closures occur. Hybrid quantum–classical approaches, such as QAOA implemented on IBM Qiskit, have shown the ability to generate feasible routes more quickly than classical heuristics, enabling dynamic rerouting under real-world constraints. The Capacitated Vehicle Routing Problem (CVRP), a widely studied NP-hard problem, has been identified as a strong candidate for quantum acceleration due to its high classical computational demands. Research utilizing the QPIA has further shown that leveraging quantum principles such as entanglement and interference enables heuristic sampling of high-quality routing solutions, even with limited quantum hardware resources [18].

Gate-model encodings for routing. Circuit-based workflows encode routing as Hamiltonians minimized via unitary evolutions or variational layers (e.g., QAOA), enabling VRP/TSP formulations within gate-model pipelines [9], [15].

Additionally, hybrid quantum-classical approaches have been proposed to mitigate quantum hardware constraints while still achieving efficiency gains over classical methods [34]. Several studies show QAOA and quantum annealing improve VRP efficiency and enable congestion-aware dynamic routing on IBM Qiskit and D-Wave platforms, though hardware limits remain [10], [47], [48]. Additionally, quantum annealing has shown promise in real-time traffic flow optimization, enabling dynamic routing decisions based on congestion patterns, which classical approaches struggle to handle efficiently [10].

Bike-sharing and logistics deployments. Annealing-based logistics (e.g., bike-sharing rebalancing) extend naturally to fleet routing and scheduling under time-varying demand, where QUBO/Ising mappings yield tractable solutions on annealers [16].

Beyond vehicle routing, logistics optimization-a fundamental challenge in transportation networks-stands to benefit significantly from quantum computing. The exponential growth of logistical data, coupled with the complexity of scheduling, assignment, and network optimization, often renders classical solutions impractical for large-scale problems. Quantum computing offers novel approaches to tackle these challenges through quantum-enhanced metaheuristics, Grover's search algorithm, and quantum annealing. Studies have demonstrated quantum speedup in optimizing supply chain networks, fleet management, and transportation scheduling, which are core components of logistics optimization [14]. Moreover, hybrid quantum-classical strategies have been shown to outperform traditional deterministic optimization techniques in solving large-scale discrete-continuous logistics problems by integrating the strengths of mixed-integer programming with quantum-enhanced subproblems [34]. Recent research highlights how quantum annealing can outperform classical solvers in optimizing multicommodity network flows, leading to more cost-effective and efficient logistics operations [22]. Hybrid quantum-classical frameworks have been increasingly explored in supply chain management, particularly in facility location planning, fleet deployment, and inventory management, where quantum methods contribute to solving complex network optimization problems more efficiently [10]. These approaches enable real-time adaptive logistics solutions by accounting for multiple constraints simultaneously, offering significant improvements in computational efficiency and decision-making for large-scale transportation systems.

Scenario synthesis for planning. QGANs and quantum autoencoders can generate rare-event traffic scenarios (e.g., incidents, closures, extreme weather) to stress-test routing/logistics strategies and AV planning pipelines, complementing limited real-world datasets [16], [18], [49].

As illustrated in Figure 6, hybrid models have been applied to optimize the placement of EV charging stations, 5G network infrastructure, and real-time emergency evacuation routing. The workflow begins by encoding the transportation or logistics problem into a graph structure, which is then formulated as a graph optimization problem. A quantum device generates a parameterized variational quantum state to explore possible solutions, while a classical optimizer iteratively updates parameters to minimize the objective function. The final optimized quantum state is decoded back into a problem solution, such as an optimized network or routing plan. This figure highlights how hybrid models integrate quantum parallelism with classical refinement, making them suitable for large-scale, real-world applications. In transportation logistics, this enables scalable optimization under multiple constraints, bridging the gap between theoretical quantum models and deployable decision-support systems. These applications demonstrate the feasibility of integrating quantum computing into practical transportation problems, addressing challenges in scalability and solution quality within the NISQ era [50].

In the field of AV systems, quantum computing holds significant promises for enhancing perception, cybersecurity, and system verification. Quantum-enhanced machine learning models are being explored to improve AV perception by leveraging quantum parallelism for faster and more efficient pattern recognition in vast sensor data streams [14]. Quantum-assisted machine learningmodels, such as Quantum Long Short-Term Memory networks, have shown superior performance in predicting vehicle dynamics for platooning systems, enabling smoother coordination and reducing unnecessary braking and acceleration, which in turn improves fuel efficiency and traffic stability [51]. Furthermore, quantum computing presents advantages in cybersecurity by enabling post-quantum cryptography and quantum-secure communication, ensuring resilience against cyber threats in AV networks [52]. Additionally, quantum computing can improve system verification and software testing for AVs by optimizing connected subsystems and ensuring the safety of cyber-physical vehicle architectures. *AV perception and planning.* Perception and sensor-fusion accelerations via QCNNs and QSVMs translate to AV pipelines for real-time obstacle detection and trajectory planning under multi-sensor loads [14], [19], [36].

Entanglement-enabled quantum key distribution (QKD) and related networking primitives provide tamper-evident, quantum-secure channels for V2V/V2I communications [8], [14]. Given Shor's algorithm's implications for RSA/ECC, this motivates migration to post-quantum cryptography and/or quantum-secure key exchange in connected and autonomous vehicle ecosystems [23].

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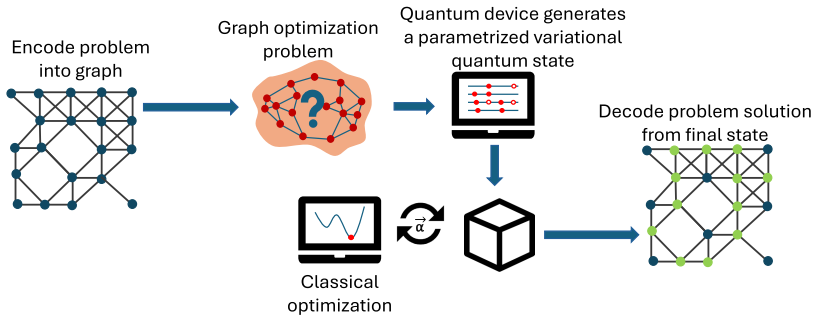


Fig. 6. Hybrid Quantum-Classical Optimization Applications, Adapted from [50].

D. Environmental Sustainability in Transportation

Traditional routing algorithms, while effective, often struggle with scalability and computational efficiency in large transportation networks. Quantum computing, leveraging quantum annealing and QAOA, provide a paradigm shift in solving these challenges. Studies such as Jagirdar [17] demonstrate the potential of quantum algorithms in optimizing traffic flow and dynamically adjusting bus routes using hybrid quantum-classical approaches. Similarly, Gautam and Ahn [18] applies the QPIA to VRPs, showing that quantum mechanics-inspired heuristics can simultaneously evaluate multiple routes, leading to improved path selection with reduced computational burden. Bentley et al. [27] further emphasizes the suitability of quantum computing for transport optimization by demonstrating significant improvements in solving the CVRP, a crucial problem in logistics and urban mobility, through novel quantum algorithmic approaches. Recent advancements in quantum-powered transport logistics have shown that QAOA and quantum annealing can enhance vehicle utilization and minimize delays while significantly improving the efficiency of VRP [53]. Additionally, applications in ITS utilize quantum computing to dynamically adjust traffic flow, reducing congestion and energy consumption in large-scale urban environments [14].

Optimizing transportation networks to minimize fuel consumption and greenhouse gas emissions is a critical step toward sustainable urban mobility. Quantum optimization techniques offer promising pathways to address this issue by improving fleet management, vehicle dispatching, and energy-efficient scheduling. Gautam and Ahn [18] highlight the application of quantum computing in reducing overall travel distances for vehicle routing, which directly contributes to lower fuel consumption and emissions. Ho et al. [54] expand on this by demonstrating how QAOA and quantum annealing can enhance logistical planning for low-carbon transportation networks, optimizing fleet operations to reduce unnecessary idling and congestion. Additionally, Hamdy et al. [55] explore the intersection of quantum computing and reinforcement learning for optimizing container movements in maritime ports, showing how intelligent quantum-assisted scheduling can minimize congestion and emissions in multimodal transport systems. Further research on quantum computing in logistics and supply chains has demonstrated its potential in optimizing transportation networks, leading to reduced fuel consumption and carbon footprints [10]. In particular, electromobility applications highlight how quantum algorithms can optimize charging schedules for battery-electric vehicles, prioritizing the use of locally generated renewable energy and significantly lowering transportation-related CO_2 emissions [56].

Beyond energy efficiency and emissions reduction, sustainability in transportation encompasses broader factors, including resource optimization, integration of renewable energy sources, and resilient infrastructure planning. Quantum computing plays a vital role in enhancing sustainability metrics by enabling advanced simulations and multi-objective optimization strategies. Ho et al. [54] explores how QML can be utilized for sustainability-focused transport planning, integrating quantum optimization models to balance energy consumption, cost efficiency, and environmental impact. Ajagekar and You [34] further discusses the intersection of Quantum AI and renewable energy integration, illustrating how quantum-enhanced optimization can improve energy distribution in transportation networks, reducing reliance on fossil fuels. Additional studies on QML applications in climate change and sustainability reinforce the potential of quantum techniques in transportation decarbonization, optimizing renewable energy distribution, and improving long-term sustainability efforts [57]. Hybrid quantum-classical frameworks applied to public transit systems have also demonstrated the ability to balance accessibility and efficiency, ensuring that transit facility allocation aligns with sustainability goals while maintaining service quality [14].

E. Evaluation and Benchmarking Protocols for Quantum Computing

To make the surveyed applications concrete and reproducible, we specify a concise validation scaffold spanning simulations/digital twins, benchmarks with classical baselines, and staged pilots. This closes the gap between algorithm proposals and deployable solutions in transportation.

1) Simulation and Digital Twins: City- and corridor-scale digital twins provide risk-free environments for offline experimentation and online what-if decision support [50]. Quantum optimizers (e.g., QUBO/annealing) can be used to establish traffic/network simulators (e.g., SUMO), to enable controlled stress tests (incidents, demand surges, weather) and closed-loop control evaluation. U.S. Department of Transportation (USDOT) provides some guidance on digital-twin-backed

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demonstrations as near-term translational pathways [50]. Large-scale simulation studies demonstrate how annealing can be exercised before field trials-for example, global signal control on a 50×50 grid and congestion minimization using Beijing T-drive trajectories [14].

2) Benchmarks and Classical Baselines: Standardized datasets and side-by-side baselines are essential. For network design, QUBO formulations solved on the D-Wave Advantage quantum annealer have been compared against classical tabu search on canonical networks, establishing a clear benchmarking template [58]. For prediction and anomaly detection, common datasets include METR-LA (Los Angeles traffic speed sensors) and PEMS-BAY (California freeway sensor data); for routing and logistics, the Capacitated Vehicle Routing Problem Library (CVRPLIB) and the Traveling Salesman Problem Library (TSPLIB); for signal control, Simulation of Urban MObility (SUMO) grids or corridors with stochastic demand. Some of the common measures are Mean Absolute Error (MAE), Root Mean Squared Error (RMSE), Mean Absolute Percentage Error (MAPE), and Area Under the Receiver Operating Characteristic curve (AUROC) or Precision–Recall curve (PR-AUC) for prediction; average delay, queue length, throughput, and fairness for signal control; and total distance, vehicles used, and tardiness or service-level violations for routing.

For example, ensuring near-term demonstration priorities, the USDOT [14], [50] conducted several pilot studies. Urban corridor signal control used a SUMO model of a 6–10-intersection corridor with noisy detectors to compare QUBO/annealing and hybrid Genetic Algorithm–Quantum Mating Operator (GA–QMO) methods against max-pressure control and reinforcement learning. Key performance indicators (KPIs) included average delay, queue length, throughput, and phase-fairness, with additional sensitivity tests to demand shocks and incidents.

VRP with time windows relied on CVRPLIB/TSPLIB instances to evaluate the QAOA, quantum annealing, and the Quantum Path Integral Approach (QPIA) against Google OR-Tools Large Neighborhood Search (LNS) and MILP under identical constraints. Here, KPIs focused on total distance, fleet size, and late deliveries, with attention to quality–runtime trade-offs and ablations over circuit depth and embedding size.

Finally, secure Vehicle-to-Vehicle (V2V) and Vehicle-to-Infrastructure (V2I) link pilots employed network simulator ns-3 or hardware-in-the-loop prototypes for QKD between roadside units (RSUs) and vehicles. Evaluation metrics included key rate, Quantum Bit Error Rate (QBER), and handshake latency, with comparisons against baseline National Institute of Standards and Technology (NIST) Post-Quantum Cryptography (PQC) schemes.

3) Quantum Configuration Reporting: For reproducibility, it is conventional to report device/backend, circuit depth/layers (p for QAOA), shots, optimizer settings, noise/error-mitigation options, and (for annealers) embedding statistics (minor-embedding ratio, chain strength). It is also common practice to publish seeds, code, data splits, and enforce identical constraints/time budgets for quantum and classical runs.

4) Empirical Findings: On IBM Quantum System One, QAOA for VRP has produced feasible tours under hardware noise, with solution quality sensitive to penalty scaling and circuit depth p, and compared against strong classical baselines (OR-Tools LNS/MILP) under equal time budgets [59]. A hybrid CVRP solver coupling a QAOAnsatz with column generation converged faster and used fewer qubits than standard QAOA while maintaining competitive gap-to-best-known, indicating near-term practicality [60]. For signal control, D-Wave annealing on a 50×50 urban grid outperformed local control and simulated annealing (lower average delay/queues across demand regimes) [61]; a real intersection study (QUBO and C-QUBO) similarly reduced congestion versus fixed-cycle timing [39]. In perception pipelines, a QDCNN achieved 0.0049 s per-frame inference-faster than thresholding (0.03 s) and chromaticity methods (1.47 s)-with improved robustness to rain, supporting AV workloads [60]. Quantum-assisted clustering on Toronto bike-share data (D-Wave backend) produced station groupings aligned with operational patterns and complementary to classical clustering for rebalancing [62]. For forecasting, hybrid quantum–classical models with data re-uploading reported up to ~5% higher accuracy than deep learning baselines with fewer epochs [62]. Finally, quantum amplitude estimation accelerated reliability-based pathfinding via the expected quadratic speedup over Monte Carlo, enabling faster expected travel-time evaluation under uncertainty [63].

This protocol-digital twins for safe iteration [50], standardized benchmarks with strong classical baselines [58], and USDOT-aligned pilot designs [50]-provides a practical path to validate quantum transportation methods and to compare them fairly against incumbent approaches.

5) Comparative Analysis of Quantum and Hybrid Approaches: Table II compares representative quantum and hybrid approaches for transportation optimization and prediction. Across applications, quantum-inspired and hybrid quantum–classical methods exhibit greater adaptability and more effective constraint handling than fixed or rule-based baselines, particularly in traffic signal control and routing. QUBO-based and annealing formulations achieve competitive performance through native constraint embedding, while gate-based methods such as Grover search mainly demonstrate theoretical potential under current hardware limits. Overall, the comparison indicates that near-term advances in transportation are driven by structure-aware hybrid approaches rather than standalone quantum advantage.

V. Challenges and Targeted Future Directions

A. Hardware Challenges

Quantum computing presents transformative opportunities in computational capabilities but is hindered by significant challenges. The fragility of qubits is a primary concern, as quantum states are highly sensitive to environmental disturbances such as noise, temperature fluctuations, and electromagnetic interference [67], [68]. These disturbances lead to decoherence, where qubits lose their quantum properties of superposition and entanglement, disrupting computations. Decoherence is quantified using metrics like relaxation time

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TABLE II
COMPARATIVE ANALYSIS OF REPRESENTATIVE QUANTUM AND HYBRID APPROACHES FOR TRANSPORTATION APPLICATIONS

Problem Domain and Method	Observed Performance Trend	Comparative Insight
Traffic signal control using quantum particle swarm optimization (QPSO) [42]	Reduced average delay compared to fixed-time control under dynamic traffic demand	More adaptive than rule-based methods; no demonstrated quantum advantage
Traffic assignment via quantum annealing with QUBO formulation [64]	Comparable solution quality to classical heuristics on benchmark instances	Strength lies in native constraint embedding rather than computational speedup
Vehicle routing using Grover adaptive search [65]	Theoretical complexity reduction for structured VRP instances	Demonstrates algorithmic potential; limited by circuit depth and scalability
Vehicle routing with hybrid quantum machine learning (Q-GAT + DRL) [66]	Faster convergence and parameter efficiency compared to classical GNNs	Hybrid learning shows promise under NISQ constraints
Traffic flow prediction using quantum neural networks [38]	Competitive accuracy trends on spatiotemporal traffic datasets	Early-stage results; performance gains are hardware-dependent
Intelligent transportation systems surveyed across quantum paradigms [14]	Identifies optimization and learning as near-term viable QC applications	Frames realistic expectations for near-term quantum deployment

(T1) and dephasing time (T2). In a study authors discussed how superconducting qubits often exhibit coherence times in the microsecond range (T1=30 μs, T2=60 μs), which limits the depth and complexity of quantum computations [67]. In practical terms, limited coherence directly caps circuit depth on NISQ devices, constraining algorithmic expressivity and making compilation/layout choices (mapping, routing) critical to preserve fidelity [17].

Quantum systems face significant challenges due to high error rates caused by the fragility of qubits and imperfections in hardware, which disrupt computations and limit scalability [69]. Errors typically occur during gate operations, measurements, and interactions between qubits [67], [68]. Surface codes are a widely used quantum error correction method because of its simple 2D structure with only nearest-neighbor interactions that perfectly fit with most of the quantum technologies and its high error threshold rate (approximately 1%), making them a fault-tolerant quantum computing approach [67]. While progress in error-corrected prototypes is encouraging, current logical qubit demonstrations still demand substantial physical-qubit overhead, creating an acute resource trade-off between depth, width, and error suppression in near-term machines [17].

In different studies it was discussed that fault-tolerant architectures, such as those using topological codes like surface codes, are essential for mitigating errors and ensuring reliable quantum computations [67], [70]. However, the scalability of these systems is constrained by the need for significant physical resources, often requiring multiple physical qubits to encode a single logical qubit effectively. Another study demonstrated the implementation of a fault-tolerant logical qubit, enabling error detection without collapsing the logical state [71]. It highlighted that while this approach reduces logical error rates, scalability is limited by the resources required for encoding logical qubits into multiple physical qubits and ensuring reliable hardware performance.

In addition, the limited qubit counts and sparse inter-qubit connectivity in NISQ devices pose significant scalability challenges, further restricting the ability to address complex, large-scale problems [68]. Similarly, a study discussed that universal quantum computers face constraints due to the low number of qubits, weak hardware scalability, and the lack of connectivity between qubits [9]. These limitations hinder the ability to process large-scale data and complex problems, impacting practical applications in fields like intelligent transportation. Analog/annealing platforms face related constraints: finite qubit numbers, sparse/structured couplers, and limited coefficient precision can bottleneck large QUBO/Ising embeddings and degrade solution quality when chaining is required [8], [9]. Despite these constraints, incremental hardware improvements (coherence, connectivity, calibration stability) and control-stack advances continue to expand feasible problem sizes for both circuit and annealing modalities [8], [17].

B. Software Challenges

Significant challenges have emerged with the growing development of quantum computing across various domains of science and engineering, including transportation, particularly in software and algorithm-related areas. The lack of mature quantum programming languages, difficulty debugging quantum programs, and the lack of domain-specific quantum algorithms for transportation applications limits practical adaptation in transportation engineering. These issues are explored in the following sections.

1) Lack of Mature Quantum Programming Tools: Programming languages like Qiskit, Quipper, Q#, and PyQuil are developed to write quantum algorithms that can execute on quantum hardware. They include built-in functionalities for constructing quantum circuits, specifying quantum gates, and implementing quantum algorithms [72]. Different research institutes and technology companies like Google, IBM, Microsoft, and Rigetti are leading the development of quantum libraries to enhance functionality and development capabilities.

Table III highlights the current state of quantum programming frameworks. A lack of standardization and interoperability across hardware platforms hinders the development of efficient libraries and algorithms for optimizing traffic flow, logistics, and vehicle routing. This creates additional challenges such as limited qubit connectivity, low gate fidelities, and the need for large numbers of qubits to enable error correction [73]. Moreover, transportation-focused libraries that bridge domain models (e.g., routing/network design) to

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TABLE III
QUANTUM COMPUTING PROGRAMMING FRAMEWORKS [72]

Framework	Features	Programming Languages	Hardware Compatibility
IBM Qiskit	Open-source, modular quantum programming framework. Tools for circuit simulation and IBM Quantum hardware execution.	Python, Cirq, OpenQASM	IBM Quantum devices
Google Cirq	Designed for NISQ devices, optimized for near-term quantum computing.	Python	Google's Sycamore processor
ProjectQ	Compiler framework allowing quantum programs to be compiled to various backends.	Python	IBM, Rigetti, Microsoft simulators
Microsoft Quantum Development Kit	SDK for quantum computing with Q# integration.	Q#, Python, C#	Azure Quantum, IonQ, Honeywell
Xanadu Strawberry Fields	Focused on photonic quantum computing for continuous-variable quantum systems.	Python	Xanadu's photonic processors
Rigetti Forest	Full-stack quantum framework with Quil support.	Quil, Python	Rigetti's Aspen series processors

quantum-native abstractions (circuits/QUBO) remain nascent, increasing integration effort for practitioners [27], [34].

2) Difficulty in Debugging Quantum Programs: Writing efficient quantum algorithms and programs requires new ways of thinking and innovative approaches to dealing with problems. The traditional debugging techniques are inappropriate for quantum programs as they differ completely from the traditional debugging technique of inspecting and troubleshooting a program's state at various points during the program execution and introduce concepts like superposition and entanglement obstructing the direct observation of quantum program's states [14]. Moreover, qubits are extremely error-prone, and quantum states collapse upon observation impedes the traditional debugging techniques of observing the states during program execution raising the need for the development of innovative debugging solutions capable of finding and correcting errors without the direct measurement of quantum states [14], [74]. Recent innovations such as surface codes and topological codes help preserve quantum information from decoherence and operational errors making quantum calculations more reliable [74]. Researchers have developed a new error-correcting code that significantly reduced error rates in quantum computations [75]. Operating a surface code-based quantum computer requires intimate classical monitoring and control of the physical qubits, with the control circuitry driven by classical logic increasing the complexity of quantum debugging [75]. Further enhancements are needed to improve the stability of quantum computations such that different sectors of science and engineering including transportation can benefit from stable, scalable, and efficient quantum programs and algorithms. In near-term workflows, practitioners rely on surrogate techniques-probabilistic assertions, shadow tomography, and hardware-aware simulation-to localize errors without destructive measurements, but these add runtime and modeling overhead [14].

3) Algorithm Development Bottlenecks: Building robust software tools and libraries for quantum algorithms is still in its early stages. Unlike classical computing, which offers mature optimization libraries for tasks like logistics and traffic management, quantum programming lacks comprehensive and modular libraries for practical use [27]. Developers face a steep learning curve due to the need for knowledge of both quantum physics and computer science [72].

Although quantum languages (e.g., Q#, Cirq, and Qiskit) integrate with traditional programming languages like Python and C#, understanding quantum-specific operations such as entanglement and superposition remains a significant barrier [72]. For example, in Qiskit, creating entanglement requires applying a Hadamard gate with the command `q = hadamard(q)`, which is difficult for traditional developers without quantum knowledge.

These challenges limit the development and adoption of scalable quantum algorithms, creating a poor ecosystem of quantum libraries that slows the advancement of quantum computing applications in transportation and other fields. Additionally, stochasticity in variational optimization (barren plateaus, optimizer noise sensitivity) and annealer embedding losses can erode end-to-end gains unless paired with problem-aware encodings, warm starts, and hybrid classical post-processing [9], [34]. For QML, qubit noise and shallow-depth constraints restrict model capacity and generalization, motivating compact ansätze, error-mitigation (e.g., zero-noise extrapolation), and data re-uploading strategies [14].

C. Implementation Challenges

The development and maintenance of quantum computing hardware require significant capital investment, particularly for superconducting quantum processors. Krinner et al. [76] highlight the need for dilution refrigerator infrastructure to maintain ultra-low temperatures, which adds substantial costs due to specialized components such as NbTi coaxial cables, microwave attenuators, and thermal management systems. Scaling quantum processors beyond 100 qubits further amplifies these financial and engineering challenges. Ajagekar and You [12] emphasize that quantum computing infrastructure remains economically restrictive, limiting its adoption across industries due to high development costs and energy demands. Additionally, Taranto et al. [77] discuss the fundamental thermodynamic constraints of cryogenic cooling, which, despite advancements, still require extensive resources, making cost-efficient scalability a major challenge. These scale-up costs also interact with reliability demands of transportation use-cases (e.g., 24/7 decision support), necessitating robust calibration pipelines and service-level guarantees that current platforms are only beginning to standardize.

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Quantum computing systems, especially superconducting quantum computers (SQCs), require substantial energy inputs for cryogenic cooling, leading to high operational costs and environmental concerns. Billat and Doeran [78] found that over a five-year period, the carbon footprint of an SQC ranges from 50,000 kg CO_2-eq (Belgian energy mix) to 150,000 kg CO_2-eq (global energy mix), with refrigeration systems like pulse tube cryocoolers as the primary contributors to emissions. Mukhanov [79] explored energy-efficient Single Flux Quantum (SFQ) logic, introducing ERSFQ/eSFQ to reduce static power dissipation, but power consumption challenges remain. Ewim et al. [80] further examined the growing environmental footprint of quantum data centers, emphasizing that, without intervention, their energy demands could rival major industries, highlighting the need for sustainable infrastructure and renewable energy integration to minimize the ecological impact of quantum computing. Emerging mitigations include renewable-powered cryo-plants and dynamic workload scheduling (e.g., batching anneals, off-peak calibration) to shrink energy intensity without sacrificing throughput.

The limited accessibility of quantum hardware forces many researchers to rely on cloud-based quantum computing platforms due to high costs, operational constraints, and environmental sensitivities [81], [82]. While cloud-based quantum computing expands access, challenges such as qubit stability, resource allocation, and security risks persist, requiring further advancements in cloud infrastructure. In terms of scalability, Yarkoni et al. [83] analyzed a Lisbon pilot project optimizing public transit routes with quantum computing, demonstrating real-time traffic optimization benefits but also highlighting computational resource limitations and hardware constraints. Similarly, Jagirdar [17] emphasized quantum computing's transformative potential in transportation engineering while acknowledging hardware bottlenecks and algorithmic inefficiencies that must be addressed before achieving large-scale, real-world implementation. Operationally, multi-tenant queues, device variability, and restricted native topologies can introduce non-deterministic latency and mapping overheads for time-sensitive transportation applications; hybrid offline–online strategies and latency-aware compilation are therefore advisable [10], [34].

D. Targeted Future Directions

Some of the targeted future directions are illustrated in this section.

1) Hardware Roadmap and Transport-Scale Tasks: Incremental advances in quantum hardware-including improved coherence times (T_1/T_2), two-qubit fidelities approaching 99%, increased connectivity, mid-circuit measurement and reset capabilities, and faster classical feedback-will directly expand the scale of solvable transportation problems [9], [17]. For example, benchmark datasets such as the Capacitated Vehicle Routing Problem Library (CVRPLIB) and the Traveling Salesman Problem Library (TSPLIB) can be tackled with deeper quantum variational circuits, while corridor-wide traffic signal control can benefit from more robust embeddings. Important milestones for transportation-focused hardware development include demonstrating measurable solution quality gains over strong classical baselines such as Google OR-Tools' LNS, achieving minor-embedding ratios below $2\times$ on problem instances exceeding 10^4 variables, and enabling sub-100 ms control latencies for real-time traffic simulations with the Simulation of Urban MObility (SUMO) platform.

2) Error Mitigation and Reliability Under NISQ Noise: Error mitigation strategies such as zero-noise extrapolation, probabilistic error cancellation, dynamical decoupling, and Clifford data regression can stabilize variational pipelines, while techniques like gauge averaging and calibrated chain strengths are particularly effective for reducing brittleness in quantum annealers [17]. Here, NISQ refers to the period where limited qubit counts and non-negligible noise remain key bottlenecks. Expected milestones include variance reductions of 20% or more across independent seeds, reproducible trade-offs between solution quality, runtime, and energy consumption, and standardized reporting of shots, circuit depth, embedding size, and error-mitigation ablations.

3) Algorithmic Progress Toward Domain-Aware Workflows: Advances in algorithm design should emphasize transportation-specific constraints. Warm-start or biased QAOA schemes, constraint-preserving mixers for time-window or precedence rules, divide-and-conquer embeddings, and schedule-aware penalties are promising directions for optimization problems [34]. For perception-related tasks, compact QCNNs or QSVMs tuned to traffic-camera spectra can reduce circuit width and depth requirements. Performance should be evaluated not only with classical datasets such as Solomon or Homberger vehicle routing benchmarks but also through fairness-bounded traffic signal control in SUMO and anomaly-detection metrics such as area under the ROC (Receiver Operating Characteristic) curve and precision–recall curve.

4) Tools, Benchmarking, and Reporting Standards: The field would benefit from standardized benchmarking protocols that pair public datasets and simulators with strong classical baselines under identical runtime and energy budgets. Suggested frameworks include SUMO or the Multi-Agent Transport Simulation (MATSim) for traffic operations, CVRPLIB/TSPLIB/OR-Library for routing, and open datasets such as METR-LA (Los Angeles traffic) or PEMS-BAY (California freeway sensors) for traffic prediction, alongside baselines such as OR-Tools LNS, MILP, max-pressure control, reinforcement learning (e.g., Deep Q-Networks and Proximal Policy Optimization), and graph-based neural networks such as STGCN or Long Short-Term Memory (LSTM) models [10], [34]. To ensure reproducibility, all experimental reports should disclose device/backend details, circuit depth or number of layers, number of measurement shots, optimizer type, and energy/runtime costs.

5) Digital Twins as a Pathway to Staged Pilots: Coupling quantum solvers with city-scale digital twins offers a safe and systematic way to test and validate methods prior to deployment. Potential pilots include corridor-level signal control (evaluated by delay, queue length, throughput, and fairness), depot vehicle routing with time windows (evaluated by distance, fleet size, and tardiness), and secure vehicle-to-everything (V2X) links (evaluated by QKD rates, QBER,

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and handshake latency). Each of these pilots should be benchmarked against robust classical and post-quantum cryptographic baselines to assess practical advantage [10], [17].

In sum, quantum computing for transportation remains constrained by NISQ-era hardware, software immaturity, and operational costs, but targeted hybrid workflows, problem-aware encodings, and energy/error–aware deployment can already yield incremental value on select optimization and perception tasks. Continued improvements in coherence, connectivity, and toolchains-paired with rigorous benchmarking against strong classical baselines-will determine when (and where) quantum methods deliver sustained real-world advantage.

VI. Conclusion and Future Work

Quantum computing offers a transformative computational paradigm for addressing complex and large-scale problems in transportation engineering. This survey has summarized its foundational principles, including qubits, superposition, entanglement, and quantum gates, and examined key algorithms such as Grover's search, Shor's algorithm, and QAOA. These methods have been applied to a range of transportation problems, including ITS, real-time traffic management, vehicle routing, logistics, and AV operations. Quantum optimization techniques, such as quantum annealing and hybrid quantum–classical approaches, have demonstrated potential in solving combinatorial problems like the VRP and adaptive traffic signal control. Likewise, QML models, including QCNNs and quantum clustering methods, have been investigated for traffic prediction, anomaly detection, and transportation decision-making.

Despite this promise, the widespread application of quantum computing in transportation remains constrained by current hardware limitations, including qubit coherence times, high error rates, limited scalability, and restricted qubit connectivity. Additionally, the lack of mature quantum programming frameworks and domain-specific algorithms presents integration challenges for existing transportation systems. Overcoming these barriers will require parallel progress in quantum hardware, error correction techniques, and the development of hybrid quantum–classical models that can bridge current classical methods with emerging quantum capabilities.

Future research should focus on developing transportation-specific quantum and quantum-inspired algorithms, scalable hybrid architectures, and improved hardware with greater qubit coherence and connectivity. Integrating quantum computing with digital twins and real-world testbeds will support pilot applications in dynamic routing, adaptive signal control, and infrastructure planning. Advancing these efforts will require strong interdisciplinary collaboration among transportation engineers, computer scientists, and quantum researchers to accelerate practical deployment.

Beyond these broad directions, our critical assessment of the literature highlights several underexplored but high-impact avenues. First, most current studies are limited to toy networks or simplified problem instances, and there is a lack of systematic benchmarking against advanced classical heuristics. Future work must prioritize fair and reproducible evaluations to validate whether quantum methods deliver measurable advantages. Second, problem formulations in transportation are often adapted from generic optimization templates; greater progress can be made by co-designing quantum algorithms that directly incorporate transportation-specific constraints such as fairness, resilience, and equity. Third, hybrid approaches have shown early promise, but few studies address robustness under NISQ conditions-developing error-tolerant, real-time algorithms should be a near-term priority. Fourth, integration with digital twins offers a unique opportunity: quantum-enhanced simulations can evaluate emergent mobility scenarios before field deployment, guiding infrastructure investment and policy. Finally, sustainability and equity remain peripheral in the current quantum transportation literature. Embedding these societal priorities into algorithm design will ensure that quantum-enabled ITS advances not only efficiency but also accessibility, safety, and climate resilience.

Overall, quantum computing holds significant potential to enhance transportation system safety, efficiency, equity, and sustainability. Promising applications include predictive crash analytics, cybersecurity, optimized routing and logistics, advanced materials for resilient infrastructure, and sustainable mobility solutions through improved battery design, energy efficiency modeling, and climate forecasting. Continued research and technological advancement in both algorithms and hardware will be essential to realize these opportunities and drive the next generation of quantum-enabled ITS.

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